

Hera



Artist's impression of *Hera* in orbit around the asteroid Didymos

Mission type

Didymos orbiter

Operator

European Space Agency

COSPAR ID

2024-180A (<https://nssdc.gsfc.nasa.gov/nmc/spacecraft/display.action?id=2024-180A>)

SATCAT no.

61449

Website

heramission.space (<https://heramission.space>)

Mission duration

1 year, 7 months, 21 days (elapsed)
2 years, 9 months (planned)^[a]

Spacecraft properties

Manufacturer

OHB SE

Launch mass

1,128 kg (2,487 lb)

Dry mass

350 kg (770 lb)

Dimensions

1.6 × 1.6 × 1.7 m (5.2 × 5.2 × 5.6 ft)

Start of mission

Launch date

7 October 2024, 14:52:11 UTC^[1]^[2] (10:52:11 am EDT)

Rocket

Falcon 9 Block 5 B1061-23

Launch site

Cape Canaveral, SLC-40

Contractor

SpaceX

Flyby of Deimos

Closest approach

12 March 2025, 12:07 GMT

Distance

300 km (190 mi)

Flyby of Mars

Closest approach

12 March 2025, 12:51 GMT

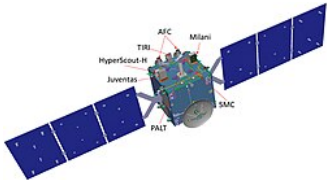
Distance

5,000 km (3,100 mi)

65803 Didymos orbiter

Orbital insertion

November 2026^[3]



Instruments



ESA insignia

Space Safety Programme

Hera is the first planetary defence spacecraft developed by the European Space Agency (ESA).^[4] Its primary mission objective is to study the Didymos binary asteroid system that was impacted in 2022 by NASA's DART spacecraft and contribute to validation of the kinetic impact method to deviate a near-Earth asteroid from a colliding trajectory with Earth.^[5] *Hera* has a mass of 1,128 kg (2,487 lb) and carries a payload of cameras, an altimeter, and a spectrometer. It is carrying two small CubeSat spacecraft, called *Milani* and *Juventas*. The spacecraft was launched on 7 October 2024 by Falcon 9.^[6] *Hera* is the first mission of ESA's Space Safety Programme.^[7]

Mission goals

Hera will measure the size and morphology of the crater created as well as the momentum transferred by an artificial projectile impacting an asteroid, which will allow measuring the efficiency of the deflection produced by the impact. It will also analyze the expanding debris cloud caused by the impact. *Hera* is intended to fully characterize the composition and physical properties of the binary asteroid system including, for the first time, the sub-surface and internal structures. It will also perform technological demonstrations linked to operations in the vicinity of a small Solar System body and the deployment of and communication with CubeSats in interplanetary space.^[5]^[8]

Planetary defense

The main objective of the *Hera* mission is to evaluate the kinetic impactor method for deflecting a near-Earth object. This planetary defense method consists of modifying the trajectory of the asteroid by launching a spacecraft at a speed of a few kilometers per second. To fulfill this objective, *Hera* will determine:

- how much momentum transfer depends on the density, porosity and characteristics of the asteroid's surface and internal structure and
- what proportion of the kinetic energy is transferred in the fragmentation and restructuring of the asteroid or in the kinetic energy of the ejected materials.^[9]

Science

Scientific objectives of the mission include exploring the properties of the two target asteroids: surface characteristics, internal porosity, and internal structure. The entire moon Dimorphos will be mapped with a spatial resolution of a few meters and the vicinity of the impact with a resolution of 10 centimeters. The mass of the moon will be estimated with high accuracy, allowing a direct estimate of the momentum transfer efficiency from DART impact.^{[10][11]}

Technology

The mission's technological objectives include the production and testing of guidance software which, by using data from several sensors, will make it possible to reconstruct the surrounding space and thus to independently define a safe trajectory around the asteroid.^[12]

Hera also carries two CubeSats which will be dropped once the asteroid is reached:^[13]

- **Juventas** will carry out subsurface and internal structure measurements, contribute to the determination of the gravity field, and provide information on the mechanical response of the surface when landing on Dimorphos
- **Milani** will collect spectral data from the surface of the two asteroids to study their surface composition and identify the presence of dust in the surrounding space



Illustration of *Hera* performing close proximity operations at Didymos



ESA-ESOC Main Control Room with the *Hera* team assembling



Testing of *Hera*'s vision-based navigation software with 3D printed asteroid models

Timeline

AIDA, the joint project with NASA

AIDA was the first operational program whose objective is to test a method of deflecting near-Earth asteroids. It was set up in 2013 jointly by scientists supported by NASA and ESA. Its objective was to test the use of an impactor-type device to deflect an asteroid that might strike the Earth. This program provided for the launch to the binary asteroid 65803 Didymos of two spacecraft: the DART impactor developed by NASA responsible for crashing at high speed on the smaller of the two asteroids and the AIM orbiter developed by ESA, which would measure the effects of the impact.^[14]

After an evaluation phase in the two space agencies, ESA decided at the end of 2016 to abandon the development of AIM due to lack of sufficient financial support from member states.^{[15][16]} NASA, for its part, decided to continue the development of DART. In this new context, terrestrial observatories were responsible for taking over partially the role of AIM.

The DART mission was launched on 24 November 2021 by a Falcon 9 rocket from Vandenberg Space Force Base and reached the binary asteroid 65803 Didymos on 26 September 2022, colliding with its satellite Dimorphos at 23:16 UTC at a relative speed of about 6.6 km/s.^[17] The project incorporated the Italian nano-satellite LICIACube, responsible for documenting and retransmitting the first 100 seconds of the impact.^[18]

Renaissance of the European project

In 2017, at the request of several member states of ESA, the agency resumed studies of a replacement for AIM, specifically a mission that was named *Hera* (after the Greek goddess). *Hera* is intended to fulfill all the objectives assigned to AIM that Earth-based observatories could not, using the components of AIM as much as possible.^[16] *Hera* was planned to be launched in October 2024 to catch a workable planetary/asteroid alignment and study the effects of the DART impact on Dimorphos, the satellite of Didymos, 4 years after it occurred.

The *Hera* mission was subsequently approved by the ESA Ministerial Council in November 2019.^[19] In September 2020, ESA awarded the construction of the spacecraft to a consortium of companies led by OHB, under a contract of 129.4 million euros.^[15] It also formalized the scientific team of the mission, made up of a principal investigator, scientific council, and four working groups^[20] covering all aspects of the mission and the scientific managers of the instruments.

Construction and testing

Hera's Core Module and Propulsion Module have been connected at OHB Bremen in Germany in August 2023.^[21] In November 2023, the spacecraft has completed acoustic testing at ESTEC in the Netherlands.^[22] The *Milani* CubeSat, integrated in *Italy*, and the *Juventas* CubeSat, integrated in

Denmark, were delivered to ESTEC for integration with *Hera* in March 2024.^{[23][24]} The spacecraft moved to final testing in March 2024.^[25] In September 2024, *Hera* was transported to Florida by Antonov Airlines's AN-124 cargo aircraft.^[26]

Launch

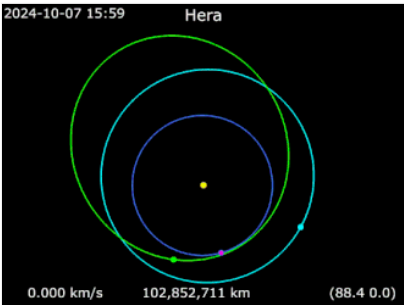
Hera launched on 7 October 2024 at 14:52 UTC by a Falcon 9 rocket lifting off from Space Launch Complex 40 (SLC-40) at Cape Canaveral Space Force Station.^[28] Usually, Falcon 9 launches for beyond geostationary transfer orbit were made from the nearby Launch Complex 39A (LC-39A) at Kennedy Space Center due to SLC-40's focus on Starlink launches. However, LC-39A was in the midst of preparing for the launch of NASA's Europa Clipper on a Falcon Heavy later in the week, which locked *Hera* into using SLC-40. Additionally, the launch occurred while Falcon 9 was grounded due to a mishap with a second stage deorbit burn happening the SpaceX Crew-9 mission the previous month. As *Hera* was to be put into a heliocentric orbit however, the Federal Aviation Administration waived the grounding for its launch as an exception. The booster used for the launch was expended to help with the trajectory; it previously flew 22 times, including on missions like SpaceX Crew-1 and the IXPE.^{[29][30]}

Cruise towards Mars

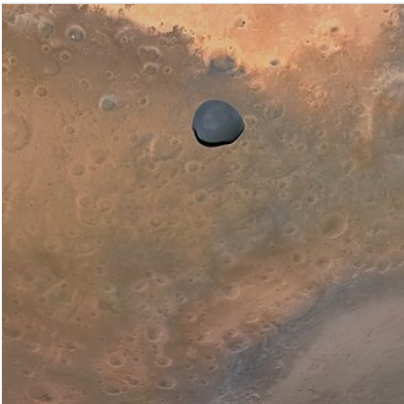
On 10 and 11 October 2024, *Hera* tested three of its science instruments (AFC, TIRI, and HyperScout H) by using them to take pictures of Earth and the Moon from a distance of more than one million km.^[31]

In October 2024, the two CubeSats aboard *Hera* were briefly activated to confirm their nominal status and test their ability to communicate with Earth. This marked the first operation of CubeSats in deep space by ESA.^[32]

Following three successful test burns performed after launch, the spacecraft conducted its first deep-space maneuver on 27 October 2024, firing its three orbital control thrusters for 100 minutes and changing its velocity by ~146 m/s, and second on 6 November 2024, firing for 13 minutes and changing its velocity by ~20 m/s.^[33]



Animation of Hera around Sun
Hera · Sun · 65803 Didymos · Mars · Earth



Deimos over Terra Sabaea, photo taken by Hera on 12 March 2025, processed by Andrea Luck.

Mars flyby

Hera performed a gravity assist at Mars on 12 March 2025, coming at a distance of 5,000 km (3,100 mi) from the surface. As part of the flyby, the spacecraft spent some time observing the Martian moon Deimos, imaging it from distance of 1,000 km (620 mi) and having its closest approach at 300 km (190 mi) away.^[34] During the flyby, *Hera* also autonomously locked onto impact craters and other surface features on Mars to track them over time, in a full-scale test of the autonomous navigation techniques it will use to navigate around its targets.^[35]

Cruise through the main asteroid belt

On 11 May 2025, the mission has captured images of the asteroid (1126) Otero from a distance of approximately 3 million km and on 19 July 2025 of the asteroid (18805) Kellyday from a distance of approximately 6 million km in order to perform instrument tests and to demonstrate agile spacecraft operations useful for planetary defence.^[36] In October 2025, ESA announced that *Hera* was on track to arrive at Didymos in November 2026—a month earlier than originally planned.^[37] Between 25 October and 1 November 2025, *Hera* was passing through the cometary tail of the third known interstellar object, the comet 3I/ATLAS. The spacecraft flew 8.2 million km from the tail's central axis.^{[38][39]} In March 2026, *Hera* successfully performed its second deep-space maneuver, burning 123 kg of hydrazine and changing its velocity by 367 m/s. This was the mission's largest maneuver in terms of fuel consumption.^{[40][41]}

Arrival at Didymos

The spacecraft will reach the binary asteroid (65803) Didymos in November 2026, four years after DART,^[37] to begin six months of investigation. *Hera* will be the first to make a rendezvous with a binary asteroid. Once close to the double asteroid, five stages will follow:

Mission phases ^[42]			
Phase	Mission phase	Duration	Mission
1	Early Characterization	6 weeks	The global shape and mass/gravity, as well as thermal and dynamical properties, of both asteroids will be determined
2	Payload Deployment	2 weeks	The release of the two CubeSats
3	Detailed Characterization	4 weeks	Meter-scale mapping of the asteroids and determination of thermal, spectral, and interior properties
4	Close Observation	6 weeks	High-resolution investigations of a large fraction of the surface area of Dimorphos, including the DART impact crater
5	Experimental	6 weeks	Morphological, spectral, and thermal properties of Dimorphos

Spacecraft

The main bus of *Hera* has a box-shape based on a central tube and adapter cone of 1.6 × 1.6 × 1.7 meters. Two solar panel wings extend from opposite sides, and a high-gain dish antenna is mounted on one face. Total launch mass of the spacecraft is approximately 1214 kg, the dry mass is 696 kg. Spacecraft deployed dimensions are 2.2 × 11.4 × 2.2 meters. The solar panels have an area of about 13

square meters. The spacecraft will use 712 W at the nominal 2.4 AU distance. Bi-propellant chemical propulsion is used for 16 orbit control thrusters and 6 reaction control thrusters, all 10-N motors. Total available delta-V is about 1300 m/s. Communications with the ground are X-band (~8.4 GHz), with two low gain antennas in addition to the high-gain dish. S-band communications (~2.2 GHz), using patch antennas, will be used to communicate with the two CubeSats named *Juventas* and *Milani*, with a range of 60 km. Spacecraft orientation is maintained by 4 reaction wheels, gyroscopes, using star trackers and solar sensors, as well two Asteroid Framing Cameras (AFC). Attitude guidance is through the Planetary Altimeter (PALT).^{[43][44][45]}

Scientific instruments

Asteroid framing cameras (AFC)

The main instruments of *Hera* are the two AFC cameras (Asteroid Framing Cameras), developed by the company JenaOptronik. Identical and redundant, they each have a FaintStar panchromatic sensor of 1020 × 1020 pixels with a telephoto lens. The field of view is 5.5 × 5.5 degrees, and the spatial resolution reaches one meter at a distance of 10 kilometers. These cameras are to provide physical characteristics of the surface of the asteroid Didymos and Dimorphos as well as the crater created by DART and the Juventas landing zone.^{[10][46][47][48]}

Hyperspectral imager – HyperScout-H

HyperScout-H is a hyperspectral imager that must provide images in a spectral range between 650 and 950 nm (visible and near infrared). The instrument makes its observations in 25 distinct spectral bands. It is developed by *cosine Remote Sensing*. This is a specific version developed for *Hera*, different from the standard HyperScout.^{[49][50][51]}

Planetary altimeter (PALT)

PALT is a micro-Lidar planetary altimeter using a laser emitting an infrared light beam at 1.5 microns. Its track on the ground is 1 meter at an altitude of 1 kilometer (1 milliradian). The altitude measurement accuracy is 0.5 meters. Its frequency is 10 Hertz.^[52]

Thermal Infrared Imager (TIRI)

TIRI is a thermal infrared imager provided by the JAXA, the Japanese space agency. The spectral range observed is between 8 and 14 microns and it has 8 filters. Its visual range is 13.0 × 9.9°. The spatial resolution is 0.013° per pixel.^{[53][54]}

X-Band Radio Science (X-DST)

The mass of the two asteroids making up the binary system, the characteristics of their gravity field, their rotational speed, and their orbits will be measured using radio wave disturbances caused by the Doppler effect. The measurements relate to the radio exchanges between *Hera* and Earth stations but

also between *Hera* and the CubeSats. Due to the low orbit in which the CubeSats will circulate, these last measurements are crucial to determine the gravity of Didymos.^[55]

Main characteristics of the instruments^[56]

	AFC	HyperScout-H	PALT	TIRI
Type	Visible Imager	Spectral imager	Altimeter	Thermal infrared imager
Mass (kg)	<1.5	5.5	4.5	<4.4
Visual range (degrees)	5.5	15.5 × 8.3	non-applicable	13.3 × 10
Spatial resolution (microradians)	94.1	133	1000	226
Spectral band (nanometers)	350-1000	665-975		700-1400
Others		25 spectral bands	vertical precision : 0.5 m.	6 filters
Power (Watts)	<1.3	2.5 (average) - 4.5 (peak)	<14.5	20 (average) - <30

CubeSats

Two CubeSat type nanosatellites, named *Milani* and *Juventas*, are transported by *Hera* and released before arrival in the asteroidal system (65803) Didymos. They are responsible for carrying out investigations that complement those of their carrier ship. Both CubeSats are built around a similar platform. These are 6U-XL CubeSats with a mass (including propellant) of approximately 12 kilograms. They are 3-axis stabilized and have a cold gas propulsion system. They communicate with the mothership in S-band. The Doppler effect affected radio link is used to measure the characteristics of the gravitational field of the binary system. They have a visible light camera and star trackers which are used to determine the dynamic variations of Didymos. Finally, the two CubeSats are equipped with accelerometers which will be used to determine the properties of the surface of Dimorphos if the CubeSats land on its surface as planned at the end of their mission. *Juventas* was developed by GomSpace and *Milani* by Tyvak International.^{[57][58][59][60][61]}

Juventas

The CubeSat *Juventas* aims to determine the geophysical characteristics of Dimorphos. The probe will map its gravity field and determine its internal structure as well as the characteristics of its surface.^[62] To fulfill these objectives, it carries the following instruments:

- The JuRa radar operating in the 50–70 MHz frequency with a spatial resolution of 10 to 15 meters. It is the first instrument to probe the inner layers of an asteroid.^[63] It uses two dipole antennas with each branch measuring 1.5 meters. Each measurement session can last up to 45 minutes. It occupies a volume of less than 1U and its mass is less than 1300 grams.^[62]
- The GRASS gravimeter.^[64] whose dynamic range is 5 × 10^{−4} and sensitivity is 5 × 10^{−7}. Its mass is less than 380 grams.^[65]
- A camera.^[62]

- The radio link with the mother ship (measurement of the Doppler effect).^[55]

Milani

The CubeSat *Milani* (named after Andrea Milani) aims to take images and measure the characteristics of the possibly present dust. It will map the two asteroids forming the binary asteroid 65803 Didymos, characterize their surface, evaluate the effects of the DART impact, contribute to the measurements of the gravitational field of the asteroids, and determine the characteristics of the dust clouds possibly located around the asteroids.^[23] To fulfill these objectives, it carries two instruments:^[58]

- The ASPECT hyperspectral imaging spectrometer is the main instrument. It works in visible and near infrared light (0.65 to 2.50 microns). Its spatial resolution is 2 meters at 10 kilometers and its spectral resolution is less than 40 nanometers.^[66]
- The VISTA thermogravimeter is responsible for detecting dust (5 to 10 microns), volatiles (such as water) and light organic materials.^[67]

See also

- Near-Earth Object Coordination Centre (ESA's NEOCC)
- Planetary Defense Coordination Office (NASA's PDCO)
- Double Asteroid Redirection Test (DART mission)
- List of European Space Agency programmes and missions

Notes

- a. 2 year, 3 month cruise phase, followed by a 6 month science phase

References

1. "Hera" (https://www.esa.int/Space_Safety/Hera). *ESA*. September 2022. Retrieved 20 October 2022.
2. "Falcon 9 (Block 5) | Hera" (<https://nextspaceflight.com/launches/details/2657>). *NextSpaceflight.com*. Retrieved 4 September 2024.
3. "ESA's Hera targets early arrival at Didymos asteroids" (https://www.esa.int/Space_Safety/Hera/ESA_s_Hera_targets_early_arrival_at_Didymos_asteroids). *esa.int*. ESA. Retrieved 30 October 2025.
4. "Planetary defence mission Hera heading for deflected asteroid" (https://www.esa.int/Space_Safety/Hera/Planetary_defence_mission_Hera_heading_for_deflected_asteroid). *www.esa.int*. Retrieved 6 September 2025.
5. "The Space Review: The trials and tribulations of Hera" (<https://www.thespacereview.com/article/4873/1>). *www.thespacereview.com*. Retrieved 5 September 2025.
6. Parsonson, Andrew (8 October 2024). "ESA's Hera Planetary Defence Probe Launched on SpaceX Falcon 9" (<https://europeanspaceflight.com/esas-hera-planetary-defence-probe-launched-on-spacex-falcon-9/>). *European Spaceflight*. Retrieved 5 September 2025.